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ARTICLE

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Who knows what Mary Knew? An experimental study

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ABSTRACT
This paper reports the results of an experimental philosophy study of intuitions among members of the general US population regarding a version of Frank Jackson's famous 'Mary in the black and white room' thought experiment. A large majority of respondents (82.1%, significantly above chance) reported having the intuition which is critical to the thought experiment, i.e., that Mary does not know what it is like to see red before she actually sees red. The intuition was pervasive across demographic groups.

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1. Introduction

In "Epiphenomenal qualia" (1982), Frank Jackson proposed what has since become one of the most famous thought experiments in philosophy, now widely known as "Mary in the black and white room" or "Mary the super-scientist":

> Mary is a brilliant scientist who is, for whatever reason, forced to investigate the world from a black and white room via a black and white television monitor. She specialises in the neurophysiology of vision and acquires, let us suppose, all the physical information there is to obtain about what goes on when we see ripe tomatoes, or the sky, and use terms like "red", "blue", and so on. She discovers, for example, just which wave-length combinations from the sky stimulate the retina, and exactly how this produces via the central nervous system the contraction of the vocal chords and expulsion of air from the lungs that results in the uttering of the sentence "The sky is blue". . . .
>
> What will happen when Mary is released from her black and white room or is given a colour television monitor? Will she learn anything or not? (p. 130)

Jackson took it to be "obvious" (p. 130) that Mary would learn something new about color experience upon release. In particular, she would learn what it is like to see red.[1] But she already knew all of the physical facts about

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color experience, so what she learns cannot be a physical fact. Consequently, not all facts about color experience, and therefore mental experience, are physical facts. This is the "knowledge argument" against physicalism.

Jackson's thought experiment and the knowledge argument have been extremely influential. They continue to occupy a position at the center of debates about the nature of consciousness. Entire anthologies are devoted to them (Coleman, 2019; Ludlow et al., 2004) and they are a staple of textbooks and university courses on the philosophy of mind.

As David Chalmers (2018) has observed, however, it appears that Jackson's thought experiment has not received any attention from experimental philosophers. No one has systematically investigated whether it elicits from non-philosophers the intuition that the knowledge argument builds upon. This paper reports a study which fills this gap, investigating the prevalence among the general US population of the intuition that Jackson's Mary would not know what it is like to see red before seeing red for the first time (though, for reasons explained in the Methods section below, some changes were made to Jackson's original scenario). From here on, we will refer to this intuition as the "Mary intuition".

The question of whether non-philosophers have the Mary intuition is important for two reasons. First, it might inform how we think about the knowledge argument. It certainly seems to be true that philosophers typically have the Mary intuition (including physicalists; more on this in the next paragraph). However, it may well be that philosophers would have less confidence in the intuition if it was discovered that it was shared by almost no one else in the community. Of course, it would not imply that the Mary intuition is wrong if non-philosophers tend not to have it, but it would provide a reason to seriously consider this possibility.

If philosophers' intuitions on Jackson's thought experiment are in fact highly unrepresentative, this would obviously be a concern for those who are convinced by the knowledge argument but it would also be of significance for physicalists responding to the knowledge argument. Generally, physicalists accept the intuition that Mary would not know what it is like to see red while still in her black and white room and would only learn this upon seeing red for the first time. The salient exception is Daniel Dennett (1991, 2004, 2007). What physicalists responding to the knowledge argument usually deny is that learning what it is like to see red consists in learning a new physical fact.[2] This is certainly true of the "abilities response", closely associated with David Lewis, who writes:

> knowing what it's like is not the possession of information at all. . . . Rather, knowing what it's like is the possession of abilities: abilities to recognize, abilities to imagine, abilities to predict one's behavior by means of imaginative experiments. (1983, p. 131; see also Nemirow, 1980)

On this analysis, Mary would learn, among other things, how to discriminate seeing red from seeing other colors (perhaps only black, white and shades of gray, initially) when she sees red for the first time, but she would not learn any new physical information. It is also true of the "acquaintance response", originating with Earl Conee (1994), on which Mary's learning what it is like to see red consists in becoming acquainted with a particular property of the experience of seeing red rather than in learning a new fact about the experience – in much the same way that one can become acquainted with a particular city or a particular person. And it is true of the "old fact/new guise" response, on which Mary's learning what it is like to see red consists in coming to "apprehend" facts about color experience which she already knew under a different "guise" (Horgan, 1984[3]; the "old fact"/"new guise" terminology is from; Alter (1998)). If it is the case that the Mary intuition is a peculiar and dubious response to the thought experiment common only among philosophers, then physicalists might have done better to question the intuition rather than to accept and seek to reinterpret it.[4]

Second, the question of whether the Mary intuition is shared in the general population is directly relevant to the "meta-problem of consciousness" which Chalmers has recently explored: roughly, the problem of why it is that people sometimes express puzzlement about the nature of consciousness or say that it is difficult to understand (Chalmers, 2018). Part of this is explaining why people have certain intuitions about consciousness, which Chalmers organizes in four categories:

> (1) *explanatory intuitions* holding that consciousness is hard to explain. These include gap intuitions holding that there is an explanatory gap between physical processes and consciousness, and anti-functionalist intuitions holding that explaining behavioural functions does not suffice to explain consciousness. Closely related are (2) *metaphysical intuitions*, including dualist intuitions holding that consciousness is non-physical, and fundamentality intuitions holding that consciousness is somehow fundamental or simple. There are also (3) *knowledge intuitions*: these include both first-person knowledge intuitions holding that consciousness provides special knowledge from the first-person perspective (like Mary's knowledge of what it is like to see red on leaving the black and white room), and third-person ignorance intuitions, such as the intuition that it is hard to know the consciousness of other people or other organisms (such as what it is like to be a bat). There are (4) *modal intuitions* about what is possible or conceivable, including the 'zombie' intuition that a physical or functional duplicate of us might lack consciousness, and 'inversion' intuitions, such as that someone else might be experiencing red when I experience green. (2018, p. 12; emphases original; numbering added.)

But the question of why people have these intuitions only needs to be answered if – again – there are people beyond a small group of philosophers who do in fact have the intuitions.

Chalmers expresses the belief that the relevant intuitions "are widely shared beyond philosophy" and that many of the intuitions "can be teased out of ordinary subjects" but acknowledges that "it is an empirical matter just how widespread they are" (2018, p. 13). It is in this context that he makes the observation that there does not yet appear to have been an experimental investigation into the prevalence of the intuition on which Jackson's knowledge argument depends – and not a great deal of empirical research directly relevant to the other kinds of intuitions which he itemizes.[5]

David Rosenthal takes a different view from Chalmers on the prevalence of the relevant intuitions. In a commentary on Chalmers' article, he writes that it is "questionable how widespread [such] intuitions are" (2019, p. 197) and expresses particular concern about the "teasing out" of the relevant intuitions. He holds that, insofar as non-philosophers do report the kinds of intuitions Chalmers is concerned with, "they result at least in part from background set-up, prompting, and tacit proto-theoretic assumptions" (p. 196) – and he clearly does not take these to be innocent influences.

Chalmers revisits the issue in a (2020) response to Rosenthal and other commentators. He suggests that the prevalence of the knowledge intuitions – which obviously includes the Mary intuition – may be particularly amenable to experimental investigation. Before making a prediction about the prevalence of these intuitions, he introduces the notion of "population-level" universality. It is not necessary for literally everyone, or even every normal adult, to have an intuition in order for that intuition to have population-level universality. Rather, an intuition will have population-level universality if it is shared by a majority of people in every population, where a population is presumably a group of people within some standard demographic category (Chalmers does not provide a specific notion of population). He then writes:

> Perhaps the most promising intuitions to study experimentally are knowledge intuitions. This is both because they are especially easy to state in plain language, and because they are among the more promising candidates to be universal intuitions [authors' note: it is clear from context that Chalmers is referring to population-level universality]. For example, we could ask whether Mary (who knows all about the physical processes in the brain but has never seen colours) knows what it is like to see red before she leaves the room, or whether she learns what it is like to see red when she leaves the room. I would predict strong negative and positive responses to these questions respectively. (2020, p. 242; see also Chalmers, 2018, pp. 14, Footnote 9.)

So, there are two large philosophical issues potentially illuminated by an experimental philosophy study of Mary in the black and white room: the viability of the knowledge argument itself and the very existence of a meta-problem of consciousness. In this paper, we report on a survey measuring the prevalence of the Mary intuition in the general US population and in various demographic subsets of that population. A finding that the Mary

intuition is widespread would support the viability of the knowledge argument (insofar as it would show that the knowledge argument does not exploit an intuition which is peculiar to philosophers) and provide evidence of the existence of a meta-problem of consciousness; any other finding would draw these into question.

2. Methods

2.1. Preliminary

Given the close connection of this project to their work, Chalmers, Rosenthal and Jackson were invited to comment on drafts of the survey as it was developing. Suggestions from all were incorporated but this should certainly not be interpreted as indicating that any of them endorses the form that the survey ultimately took.

2.2. Participants

Participants were sourced using Prolific, a UK-based service for recruiting survey participants primarily for academic research. The survey itself was hosted on SoSciSurvey, a German online platform for conducting surveys and questionnaires.

Participants were informed at the beginning of the survey that it was part of an experiment; that they would be anonymous; that their participation was voluntary; and that they could withdraw before completing the survey. They could only participate if they indicated their consent. Each participant who completed the survey was paid 63p.[6]

A total of 602 complete responses were collected. We applied Prolific's function for collecting a sample which is representative of the general US population with respect to age, gender, and race. As detailed below, we also collected information from participants about their level of educational attainment but the sample was not representative in this respect.

2.3. Survey design

The following demographic information was collected at the beginning of the survey:

Gender identity (female/male/other).

Age (18–27/28–37/38–47/48–57/58 or above).

Level of educational attainment (did not complete high school/completed high school/associate degree/bachelor's degree/master's degree/doctoral degree).

Race (white/mixed/Asian/Black/other; multiple selections possible).

The version of the thought experiment which was presented to participants was intended to be easily comprehensible and the science-fiction quality of the original was minimized. It was as follows:

> Mary is a brilliant scientist. She is the world's leading expert on the neurophysiology of vision. Actually, she knows all the physical information about color perception that there is to know. She learned everything in the textbooks, so she knows about how light reflects off objects and about how different wavelength combinations stimulate the retina and about the activity in the brain this causes. She also did her own experiments, so she was able to gain all of the physical information about color perception that other scientists did not already know. She knows literally every piece of physical information about color perception.
>
> However, Mary was born color-blind and she remains color-blind. She has only ever seen in black and white and shades of gray. She has seen things which look blue and red and every other color to people who are not color-blind, but they did not look like that to her. Mary saw them as black or white or some shade of gray.
>
> Because she knows everything there is to know about color perception, Mary knows that it is theoretically possible for her to have an operation which would allow her to have color vision. However, she also knows that this operation would be extremely difficult and too risky for surgeons to do. One day, however, Mary is told that recent improvements in surgical equipment and surgical techniques mean that the operation can now be done safely and that there is a surgeon who is willing to do the operation. She decides to have the operation.
>
> Before the operation, Mary decides that the first color she would like to see when she wakes up from the operation is red. She arranges for someone to give her a large red tomato as she wakes from the anesthetic. The day before the operation she works out precisely how light rays reflecting off the large red tomato will affect her visual system and her brain.

After reading the vignette, participants were asked to respond to a series of questions, arranged in three sets, as follows.

2.3.1. Knowledge pre-operation questions

The first set of questions tapped participants' intuitions on the critical issue of what Mary knew before the operation:

(1) Before the operation, Mary knows all the physical information about what will happen when she sees the red tomato, but she has not yet seen the red tomato. At this point, does she know what it is like to see red? (Yes/No/Don't know).
(2) If you would like to, you can add comments on your answer to Question 1 or the reason(s) for it. Please write no more than about 50 words.

(3) When you were answering Question 1, were you intuitively inclined to give one answer before eventually deciding to give another? (Yes/No/Don't know).

Question 4 was a manipulation check.

Participants who answered "No" or "Don't know" to Question 1 completed the remaining questions in this set (Questions 2 and 3) and then progressed to two further sets of questions regarding what Mary could imagine pre-operation and what she would learn post-operation. Participants who answered "Yes" to Question 1 completed the remaining questions in this set (Questions 2 and 3) but not the further questions. As these participants had the intuition that Mary already knows what it is like to see red pre-operation, we did not see any value in asking them about what she could imagine pre-operation and what she would learn post-operation.

2.3.2. Imagination pre-operation questions

The next set of questions tapped intuitions on what Mary could imagine pre-operation:

(5) Before the operation, Mary knows all the physical information about what will happen when she sees the red tomato, but she has not yet seen the red tomato. At this point, would Mary be able to correctly imagine what it is like to see red? (Yes/No/Don't know).
(6) If you would like to, you can add comments on your answer to Question 5 or the reasons for it. Please write no more than about 50 words.
(7) When you were answering Question 5, were you intuitively inclined to give one answer before deciding to give another? (Yes/No/Don't know).

2.3.3. Learning post-operation questions

The final set of questions was directed toward intuitions on Mary's situation post-operation:

(8) Before the operation, Mary knows all the physical information about what will happen when she sees the red tomato, but she has not yet seen the red tomato. When she wakes from the operation and she sees the red tomato, will she then learn what it is like to see red. (Yes/No/Don't know).
(9) If you would like to, you can add comments on your answer to Question 8 or the reason(s) for it. Please write no more than about 50 words.

(10) When you were answering Question 8, were you intuitively inclined to give one answer before deciding to give another? (Yes/No/Don't know).

Question 11 was a manipulation check.

All questions were required except for those which invited free-text responses (Questions 2, 6 and 9). All participants were also required to indicate whether they had had previous contact with the material in the survey.

The rationale for asking respondents at three points whether they had been intuitively inclined to answer a question in one way before ultimately deciding to answer in another is that there may be room to question whether a respondent whose final answer to a question differed from their immediate reaction was really reporting an intuition rather than a more considered response. Our view is that there is no reason that a response to a question such as whether Mary would know what it is like to see red pre-operation should not be considered an intuitive response just because it was not a respondent's immediate reaction (and we report results accordingly). It is actually not clear how one could answer any such question other than intuitively, no matter how long they considered it and how many times they changed their mind: philosophers who have been reflecting on such questions for decades can still only provide intuitive responses to them. However, we included the questions asking whether respondents changed their minds before providing responses to the first question in each set to make this information available to those who might see the matter differently and take it to be important. This was also the rationale for asking respondents if they had had previous contact with Jackson's thought experiment.

The questions in Set 2 are not directly relevant to the knowledge argument, which turns on whether Mary *knows* what it is like to see red before she wakes from the anesthetic (or leaves her room, in the original version), not whether she can *imagine* what it is like. This is clear in "Epiphenomenal qualia" (1982) and Jackson emphasized it in "What Mary didn't know" (Jackson, 1986), writing that "the knowledge argument claims that Mary would not know what the relevant experience is like. What she could imagine is another matter" (p. 295). However, it was considered that the survey would provide a valuable opportunity to source intuitions on this question from respondents who also had the intuition that Mary would not know what it is like to see red prior to doing so for the first time (or who did not know whether she would). Results would potentially be instructive regarding folk intuitions on the relationship between knowledge and imagination.

3. Results

Of the 602 complete responses, 8 (1.3%) were excluded from analysis. Four were excluded for failed manipulation checks. One was excluded because the respondent had participated in a pilot study. Three responses were excluded because they included nonsensical or irrelevant comments.[7] We did not exclude responses from 13 participants who reported previous contact with the thought experiment or from 3 participants who were not sure whether they had had previous contact (2.7% of final sample, too small to analyze independently or to influence overall results). Consequently, 594 cases were included in the analysis. The excluded group is negligibly small compared to our effect sizes, so our conclusions are likely robust to alternative exclusion criteria.

In the following, we report the frequency of responses to the multiple choice (Yes/No/Don't know) questions as well as the results of exact binomial, Chi-squared and point biserial correlation tests conducted to determine if certain responses were favored by statistically significant majorities and if responses varied from a chance distribution. Although we did not have specific predictions about variation between demographic groups, we also ran statistical tests to investigate differences regarding responses to Question 1, which was the critical question in determining whether respondents had the Mary intuition. We report using uncorrected p-values throughout. Unavoidably, some subjective judgment was involved in determining which of the responses to the questions which allowed free-text responses to report. We provide examples of responses which, we take it, represent noteworthy patterns.

We report the results of all of the statistical tests which we conducted.

3.1. Knowledge pre-operation responses

Question 1 was the crucial item asking respondents if Mary knows what it is like to see red before she has the operation and sees the red tomato. Answering "No" to this question amounts to reporting the Mary intuition.

In response to Question 1, 82.1% of the 594 respondents whose answers were analyzed answered "No". Of the remainder, 16.0% answered "Yes" and 1.9% answered "Don't know". An exact binomial test revealed that a significant majority (more than 50%) demonstrated they had the Mary intuition by answering "No", rather than "Yes" or "Don't know", $p < .001$. Examining if responses differed from the distribution expected by chance, a Chi-squared test revealed that the frequency for "No" responses was significantly higher ($Z = 20.61$, $p < .001$), and the frequency for "Yes" (Z

= −7.32, $p < .001$) and "Don't know" responses was significantly lower (Z = −13.29, $p < .001$), than would be expected by chance, $\chi^2(2, N = 594) = 654.94$, $p < .001$.

Demographic analyses were carried out on the responses to Question 1 to investigate whether the Mary intuition was "widely shared" in the sense of having something approaching population-level universality. That is, we examined if a majority of respondents in every demographic group had the Mary intuition. We also sought to detect any demographic variation. Those who responded "Don't know" to this question were excluded from analysis, as the group was too small (11 respondents) for meaningful statistical characterization, once further divided into demographic groups.

While a majority of both genders had the Mary intuition, the intuition was more prevalent among women than men. An exact binomial test for each gender revealed that both a significant majority of men and a significant majority of women had the Mary intuition, with $p < .001$ in both cases. A Chi-squared test revealed that women were significantly more likely to answer 'No' (88.0%) than 'Yes' (12.0%) to Question 1 ($Z = 2.80$, $p = .004$) than men were ('No' = 79.2%, 'Yes' = 20.8%; $Z = -2.80$, $p = .004$), $\chi^2(2, N = 583) = 8.15$, $p = .017$. Although too small a sample for inferential statistics, of the five individuals who reported their gender as 'Other', 80% responded to Question 1 with 'No'; 20% responded with 'Yes'.

Exact binomial tests were conducted on responses for all education levels (expect for those who did not complete high school, a group of only two); they revealed that a significant majority of respondents at each education level had the Mary intuition, $p < .05$ in all cases. A point-biserial correlation analysis showed that respondents with a higher level of education were significantly more likely to answer 'Yes' than 'No' than those with a lower level of education, $r(581) = .104$, $p = .012$. That is, more educated individuals were less likely to have the Mary intuition than less educated individuals (see Table 1).

A series of exact binomial tests revealed that a significant majority within each age group had the Mary intuition, $p < .001$ in all cases. A point-biserial correlation analysis showed that older respondents were significantly more

Table 1. Level of educational attainment and responses to question 1.

		Did not complete high school	High school	Associate degree	Bachelor's degree	Master's degree	Doctoral degree
No	Count	1	135	73	173	93	13
	%	50.0%	90.0%	86.9%	81.2%	78.8%	81.3%
Yes	Count	1	15	11	40	25	3
	%	50.0%	10.0%	13.1%	18.8%	21.2%	18.8%
	Total Count	2	150	84	213	118	16

Table 2. Age and responses to question 1.

		18–27	28-37	38-47	48-57	58 or above
No	Count	85	96	72	87	148
	%	81.0%	80.0%	73.5%	91.6%	89.7%
Yes	Count	20	24	26	8	17
	%	19.0%	20.0%	26.5%	8.4%	10.3%
	Count	105	120	98	95	165

likely to answer 'No' than 'Yes' to Question 1 than younger respondents, $r(581) = -.118$, $p = .004$. That is, older respondents were more likely to have the Mary intuition than younger respondents (Table 2).

A series of exact binomial tests revealed that a significant majority within each racial group had the Mary intuition, $p < .05$ in all cases. A Chi-squared test found significant variation in responses to Question 1 according to race, $\chi^2(4, N = 575) = 13.05$, $p = .011$. When compared to other groups, individuals who identified as white responded 'No' more frequently and individuals who identified as Black responded 'No' less frequently (Table 3).

In summary, a large majority of respondents in each demographic group had the Mary intuition, with the exception of respondents who did not complete high school, for which the sample size was only 2.

A total of 334 respondents took the opportunity to add comments on the answer given to Question 1 or the reasons for that answer (setting aside 12 who gave answers such as "na" or "none"). These responses were not highly informative, with the vast preponderance effectively restating their answer to Question 1 or providing only minimal elaboration of it. Some examples follow, with the answer that the relevant respondent provided to Question 1 in brackets[8]:

(1) "She may not have seen red herself, but she understands the neurological process of seeing red better than almost anyone else in the world. Therefore, she knows what it is like seeing red, even though she hasn't experienced it herself" ("Yes".)

Table 3. Race and responses to question 1.

		White	Black	Asian	Mixed	Other
No	Count	340	62	43	18	17
	%	85.9%	70.5%	84.3%	85.7%	89.5%
Yes	Count	56	26	8	3	2
	%	14.1%	29.5%	15.7%	14.3%	10.5%
	Count	396	88	51	21	19
Z		2.30 ($p = .022$*)	−3.60 ($p < .001$*)	.20 ($p = .866$)	.30 ($p = .779$)	.70 ($p = .474$)

Note: Although it was possible in the survey to make multiple selections, those who made multiple selections including "Mixed" were only included in the "Mixed" category for purposes of analysis. The reason is that the number of individuals who selected "Mixed" plus any particular subset of races was too small for analysis. Additionally, eight individuals were removed from this analysis because they did not report their race consistently (in particular, they indicated that they belonged to multiple races but did not also indicate being of mixed race).

(2) "Because of all that she knows, she knows what it is like to see red even though she hasn't before." ("Yes".)
(3) "Although she hasn't physically seen the red color yet, because of her knowledge of neurophysiology of vision, she understands what people who do actually see red experience and feel during that time." ("Yes".)
(4) "She may understand the physiology behind seeing the color red, but she has never experienced it for herself and, therefore, doesn't know what it's like." ("No".)
(5) "She can describe what physically happens to make a person see red, but she doesn't know what it feels like to experience it." ("No".)
(6) "She knows the physical reasons for colors, but she has never experienced actually SEEING the color red. So she doesn't know what it's like – at least from a personal experience." ("No".)
(7) "She does know how seeing the color red will affect her brain, but she has not personally had this experience yet." ("No".)
(8) "Mary knows theoretically about what it is like to see red, but does not have any experience of seeing red or any other color to have a realistic frame of reference to know or understand subjectively or directly what seeing red is actually like." ("No".)

Question 3 asked respondents if they were intuitively inclined to give one answer to Question 1 before deciding to give another. Of the respondents, 68.5% answered "No"; 28.8% answered "Yes"; and 2.7% answered "Don't know" ($N = 594$). There were more "No" responses ($Z = 14.85$, $p < .001$), and fewer "Yes" ($Z = -1.92$, $p = .055$, marginally significant) and "Don't know" ($Z = -12.93$, $p < .001$) responses, than would be expected by chance, $\chi^2(2, N = 594) = 391.59$, $p < .001$.

We also analyzed the responses to Question 3 among only those who had reported the Mary intuition. Of this group, 73.4% answered Question 3 with "No"; 24.1% answered "Yes"; and 2.5% answered "Don't know" ($N = 488$). There were more "No" responses ($Z = 15.32$, $p < .001$), and fewer "Yes" ($Z = -3.50$, $p < .001$) and "Don't know" ($Z = -11.81$, $p < .001$) responses, than would be expected by chance, $\chi^2(2, N = 488) = 386.38$, $p < .001$.

3.2. Imagination pre-operation responses

The "Imagination Pre-Operation Questions" were only put to those who answered Question 1 with "No" (i.e., who reported having the Mary intuition) or "Don't know", a group of 499.

Question 5 asked if Mary would be able to imagine correctly what it is like to see red before her operation. A Chi-squared test on the responses showed that significantly more respondents answered "No" (60.1%; $Z = 10.36$, p

< .001), and significantly fewer answered "Yes" (23.9%; $Z = -3.67$, $p < .001$) or "Don't know" (16.0%; $Z = -6.69$, $p < .001$), than would be expected by chance, $\chi^2(2, N = 499) = 165.70$, $p < .001$.

A total of 239 took the opportunity to add comments on the answer given to Question 5 or the reasons for that answer (setting aside 8 who gave answers such as "na" or "none"). Several indicated that the word, "correctly", in Question 5 caused some doubt. Some examples follow, with the answer that the relevant respondent provided to Question 5 in brackets:

(1) "Based on description, she can imagine red in theory. However, she does not have the ability to confirm that her imagination is correct." ("Yes".)
(2) "Maybe, she can definitely imagine what its like to see red, but imagining correctly is another story." ("Yes".)
(3) "She would be able to deduce an approximation of what she might see but not exactly." ("Yes".)
(4) "Mary can imagine what red may be like, but without ever seeing color she cannot correctly imagine what red actually looks like with normal vision." ("No".)
(5) "I'm sure she's thought about it and tried to imagine it many times – but I changed my answer because I saw you said CORRECTLY. I think It's almost impossible for her to correctly imagine what color is like." ("No".)
(6) "I would have answered 'to some degree' but it wasn't an option. Mary may be able to imagine what it is like to see red, but how could she imagine something if she never experienced it? She may imagine the color red based on her academic study, but that would involve imagining how others perceive red." ("No".)
(7) "If I was told a description of a new animal, I could imagine it but it may not be correct." ("No".)
(8) "She can imagine what the color would be like but not 'correctly' since she's never seen it before." ("No".)
(9) "Her imagination would just be a guess, with no guarantee that it's correct." ("No".)
(10) "Mary is able to imagine what the red tomato will look like before her surgery. However, her image of what that red tomato will look like will not be 100% correct." ("Don't know".)
(11) "I don't know the answer to this. She would certainly be able to imagine what seeing color is like, but whether she could do this correctly or not is impossible to know." ("Don't know".)

In response to the question of whether they were intuitively inclined to give one answer to Question 5 before deciding to give another, 74.0% of respondents answered "No"; 23.2% answered "Yes"; and 2.8% answered "Don't know". There were more "No" responses ($Z = 15.71$, $p < .001$), and fewer "Yes" ($Z = -3.90$, $p < .001$) and "Don't know" ($Z = -11.81$, $p < .001$) responses, than would be expected by chance, $\chi^2(2, N = 499) = 401.68$, $p < .001$.

3.3. Learning post-operation responses

The "Learning Post-Operation Questions" were only put to those who answered "No" or "Don't know" to Question 1 (the same group who were asked the "Imagination Pre-Operation Questions").

Question 8 asked if Mary would learn what it is like to see red when she wakes from the operation and sees the red tomato. A Chi-squared test conducted on the responses found that significantly more individuals answered "Yes" (90%; $Z = 21.92$, $p < .001$), and significantly fewer answered "No" (5.2%; $Z = -10.88$, $p < .001$) or "Don't know" (4.8%; $Z = -11.04$, $p < .001$), than would be expected by chance, $\chi^2(2, N = 499) = 720.56$, $p < .001$. An exact binomial test showed that a significant majority answer "Yes", rather than "No" or "Don't know", to Question 8. We expected that there would be a high correspondence between respondents who reported the Mary intuition and respondents who indicated, in response to Question 8, that Mary would learn what it is like to see red when she wakes from the operation and sees the red tomato.[9] A Chi-squared test revealed that, among this group, significantly more individuals responded "Yes" (90.6%; $Z = 21.90$, $p < .001$), and significantly fewer answered "No" (5.9%; $Z = -10.87$, $p < .001$) or "Don't know" (4.5%; $Z = -11.03$, $p < .001$), than would be expected by chance, $\chi^2(2, N = 488) = 719.53$, $p < .001$. An exact binomial test revealed that a significant majority of those who had the Mary intuition responded "Yes" to Question 8, $p < .001$.

A total of 200 participants provided comments on the answer given to Question 8 or the reasons for that answer (setting aside 10 who gave answers such as "na" or "none"). It was notable that, among participants who had reported the intuition that Mary would not learn what it is like to see red, several expressed doubt as to whether the change that Mary would undergo is appropriately described as an instance of learning and indicated that they thought the word "experience" would be more appropriate. Some examples follow:

(1) "I think the correct word is she will UNDERSTAND or EXPERIENCE what it is like to see red."

(2) "I would not say that she has learned what it is like to see red. I would say that she has now experienced the color red."
(3) "I don't think that this is learning- she already knows a great deal about what it is to see red. She will experience the redness of red for the first time".
(4) "I don't know that "learn" is the right word. 'Experience' would be better."
(5) "I like the word experience more than learn."

Among those who answered "Don't know" to Question 8, several commented that whether Mary would learn what it is like to see red upon waking from her operation and seeing the red tomato would depend on whether the operation was successful. Some examples follow:

(1) "I would say it would all depend on the success of the operation."
(2) "If the operation worked than she will. There is no knowing if it has or has not because the passage did not tell us."
(3) "Not sure, it completely depends if the operation was a success or not, or if she still shes black/grey then it isnt truly red. Depends on outcome of operation."

Question 10 asked respondents if they were intuitively inclined to give one answer to Question 8 before deciding to give another. Of the respondents, 82.2% answered "No"; 14.4% answered "Yes"; and 3.4% answered "Don't know" ($N = 499$). More respondents answered "No" ($Z = 18.89$, $p < .001$), and fewer answered "Yes" ($Z = -7.31$, $p < .001$) or "Don't know" ($Z = -11.58$, $p < .001$), than would be expected by chance, $\chi^2(2, N = 499) = 544.53$, $p < .001$.

We also analyzed responses to Question 10 among only those who had reported the intuition that Mary would learn what it is like to see red when she wakes from her operation and sees red for the first time. Of this group, 86.0% answered Question 10 with "No"; 10.7% answered with "Yes"; and 3.3% answered with "Don't know" . More respondents answered "No" ($Z = 19.32$, $p < .001$), and fewer answered "Yes" ($Z = -8.31$, $p < .001$) or "Don't know" ($Z = -11.01$, $p < .001$), than would be expected by chance, $\chi^2(2, N = 449) = 563.42$, $p < .001$.

4. Discussion

4.1. The Mary intuition

The results provide strong evidence that the Mary intuition is widespread, with an overall prevalence among respondents of 82.1%. They also provide strong evidence that the intuition has population-level universality. There was no demographic group in which the proportion of respondents who had the Mary intuition was below 70% (setting aside the group who did not finish high school, for which the sample size was 2).

There was found to be variation between demographic groups. Women, older individuals, those with lower educational achievement, and those identifying as white were more likely to have the Mary intuition than the other groups in each category. Those who identified as Black were relatively less inclined than other ethnicities to have the Mary intuition (although a 70% majority still did).

The overall results indicate that the Mary intuition is not by any means unique to philosophers. Physicalists responding to it would not have gained anything if they had sought to show that the many philosophers who report having the intuition are in fact reporting something very unusual. That said, that 16% of respondents thought that Mary would know what it is like to see red pre-operation also shows that Dennett-style (1991, 2004, 2007) denial of the Mary intuition is not entirely rare. The results also provide evidence for a key premise on which the meta-problem of consciousness is based: overall, people do in fact have certain intuitions about consciousness which need to be explained.

It has been suggested to us that there may have been an even higher endorsement of the Mary intuition if a particular locution had not been used in the vignette, viz., "[Mary pre-operation] knows everything there is to know about color perception". The concern is that knowing everything there is to know about color perception might be taken to imply knowing what it is like to see red. What Mary knows pre-operation, of course, are all the physical facts about color experience; the question at issue is whether this is sufficient to know what it is like to see red. Accordingly, the phrasing might have influenced some respondents to answer "Yes" to Question 1 (i.e., to indicate the intuition that Mary already knows pre-operation what it is like to see red).[10]

This point is legitimate, but we believe that changing the phrasing would have had only minor consequences for our results, given the context in which the phrase occurs. The first paragraph of the vignette expresses the key point more accurately twice: "she knows all the physical information about color perception that there is to know" and "[s]he knows literally every piece of physical information about color perception". We expect that the less accurate locution will have been read in light of these, so that the intended meaning will have been clear to most.

Possibly, some resistance to the conclusion that the Mary intuition is widespread could be mounted on the basis that 24.1% of those whom we have described as having the Mary intuition, (i.e., those who answered "No" to the crucial Question 1), indicated that they had initially been intuitively inclined to answer Question 1 differently. One might argue that these respondents did not really report the Mary intuition but, rather, a more considered view, given that their answers indicate that they gave the question some thought. As indicated in the Methods section, we do not believe that a response to a question such as Question 1 should not be considered an intuitive response just because it was not the respondent's first reaction. The final response which was provided was a report of the respondent's intuition at the time that they answered the question. In any case, given the strength of the findings overall, and given that 73.4% of those whom we have described as having the Mary intuition indicated that they had *not* been intuitively inclined to answer Question 1 differently, such an argument alone would be weak.

4.2. Mary's imagination

One point of particular interest arises from the responses to the "Imagination Pre-Operation Questions".

Respondents were more open than might be expected to the possibility that Mary would be able to correctly imagine what it is like to see red pre-operation. Again, only participants who had the Mary intuition or who indicated that they did not know if Mary would know what it is like to see red pre-operation were presented with the "Imagination Pre-Operation Questions". Even among this group of respondents, the proportion who had the intuition that Mary would be able to correctly imagine what it is like to see red pre-operation was 23.9%. It would be useful for any future study to ask respondents in this group why they would *not* assent to the proposition that Mary knows what it is like to see red pre-operation if they think she could already correctly imagine it pre-operation.

Certainly, there are plausible justifications. For example, one might think that Mary could produce an accurate mental image of red pre-operation – and indeed of every other color – but not know that it was an image of red and not some other color. That would seem to mean that Mary could correctly imagine seeing red but not know what it is like to see red (Kind, 2019; cf. Nanay, 2009). One of the comments quoted above is strongly suggestive of this position: "Based on description, she can imagine red in theory. However, she does not have the ability to confirm that her imagination is correct." Alternatively, it would be open to hold that, even if Mary *could* correctly imagine what it is like to see red pre-operation, she would not know what it is like to see red until she either sees red or exercises her

ability to correctly imagine seeing red. Either of these experiences would be sufficient for her to learn what it is like to see red, but she would need to have one of them before she could know what it is like to see red. If some respondents who thought that Mary would be able to correctly imagine what it is like to see red pre-operation were assuming that she did not exercise this ability pre-operation – perhaps thinking that the vignette implies that her first experience with red is the actual perceptual experience she has post-operation – this would explain why they also reported the Mary intuition.[11] In general, though, without further data, we can only speculate as to why such a considerable number of respondents had the intuition that Mary would be able to correctly imagine red pre-operation but did not also have the intuition that she would know what it is like to see red.[12]

4.3. Mary's learning experience

As indicated in reporting the results, we expected that there would be a high correspondence between respondents who had the Mary intuition and respondents who indicated, in response to Question 8, that Mary would learn what it is like to see red when she wakes from the operation and sees the red tomato. This was certainly the case, with 90.6% of respondents who answered "No" to Question 1 answering "Yes" to Question 8.

The two patterns of comments regarding whether Mary will learn what it is like to see red post-operation which were mentioned in the Results deserve comment. The responses which denied or questioned the appropriateness of the word, "learn", to describe the change Mary undergoes, suggesting that "experience" would be more suitable, certainly recall the "acquaintance response" to the knowledge argument in the academic literature. However, this may be misleading. A proponent of the acquaintance response accepts that Mary learns what it is like to see red when she sees red for the first time but holds that this learning consists in her becoming acquainted with a particular property of the experience of seeing red, rather than learning a new fact. This is very different from denying or questioning the appropriateness of the word, "learn", to describe the change Mary undergoes. So, the apparent similarity between the comments denying or questioning the appropriateness of the word, "learn", and the acquaintance response, is probably superficial.

The respondents who commented that whether Mary will learn what it is like to see red upon waking from her operation and seeing the red tomato depends on whether the operation is successful were making a legitimate point, given the wording of the vignette. There were relatively few such comments overall but it is possible that these participants would have answered the question about whether Mary would learn what it is like to see red upon waking from her operation affirmatively if they had been told

to assume the success of the operation. This would have resulted in a marginally stronger endorsement of the proposition that Mary would learn what it is like to see red upon waking from her operation and seeing red for the first time. Any future study should make it explicit that Mary's operation is successful.

4.4. "What it is like" and phenomenal consciousness

A couple of potential difficulties for the study, both relating to whether respondents could actually understand the thought experiment and the questions put to them, should be addressed. The first is that laypeople may not actually have a concept of phenomenal consciousness; the second is that, even if they do, they may not understand a locution such as "what it is like".[13]

Regarding the first, there are some studies which can be seen as providing evidence that laypeople do not have a concept of phenomenal consciousness. Sytsma and Ozdemir (2019) found that respondents were more inclined to accept that a robot could experience red than that it could experience pain. Insofar as color experience and the experience of pain are both paradigm phenomenal states, Sytsma and Ozdemir took the finding to be evidence that laypeople do not have a concept of phenomenal consciousness. For, if they did, then they should have been inclined to attribute the two states to a robot at the same rate (though Chalmers (2020) questions this interpretation of the data; see also Sytsma and Machery (2010)). With respect to our study, the concern would be that, if respondents did not even have a concept of phenomenal consciousness, then they could not properly engage with the thought experiment and the questions which were put to them.

The question of whether laypeople generally have a concept of phenomenal consciousness is extremely interesting but it does not present complications for our study. In order to understand and answer the questions we posed, respondents would need to have a concept of "what is it like to see red", but this is a much less complicated thing than a concept of phenomenal consciousness. A concept of phenomenal consciousness, after all, subsumes *many* other concepts: what it is like to see red; what it is like to feel pain; what it is like to hear middle-C. One can have many of the latter kind of concepts without having a concept of phenomenal consciousness. (What *would* be problematic for our study, of course, is evidence that laypeople do not have a concept of what it is like to see red.)

Another way of bringing out the point here is that the knowledge argument itself does not actually involve a notion of phenomenal consciousness. For the argument to work, it is not necessary that Mary learns something about phenomenal consciousness when she sees red for the first time (though maybe she does); it is only necessary that she learns what it is like

to see red. If she learns what it is like to see red, and if this new piece of knowledge is a fact, then not all facts about color experience are physical facts. That Mary learns something about phenomenal consciousness is not a premise in the knowledge argument. Accordingly, to have the intuition that the argument depends on – the Mary intuition – one does not need a concept of phenomenal consciousness. So, it is not a problem even if our respondents did not have such a concept.

The second issue relates to respondents' linguistic competence rather than their conceptual sophistication. The concern is that, whatever concepts respondents might have, they might not have understood what was meant by the phrase "what it is like", which may be less transparent than philosophers assume.

We do not believe that respondents generally had difficulty with understanding the phrase, "what is it like", but considering the issue does bring to light an interesting possibility. The basis for our confidence is that, if respondents generally had difficulty understanding the phrase, then we would expect to find evidence of this in the qualitative data. Among the more than 750 comments which respondents provided, there are *some* which indicate such a difficulty. For example, the following two comments were provided by respondents who thought that Mary did know what it is like to see red pre-operation: "She knows what it's like to see red but perceives it differently than the non color blind" and "She knows what it's like but she doesn't know how it feels". Quite plausibly, these respondents did not understand the phrase, "what it is like". As well, one respondent who reported the Mary intuition nonetheless provided this comment: "Does she know is rather subjective to a readers interpretation of that phrase". This might also indicate some difficulty with the phrase (though it might also suggest a very advanced understanding). But the number of such comments is extremely small. The comments just quoted, along with a very small number of others, really are the exceptional cases. The comments reproduced in the Results section, which do not suggest difficulties in understanding the phrase, are much more typical.

However, while we do not think that the phrase, "what it is like", was generally problematic, it is possible that different levels of understanding had some influence on one particular finding. As reported in the Results, there was statistically significant variation in the prevalence of the Mary intuition according to level of education. Respondents with a higher level of education were less likely to report the Mary intuition. The explanation for this which most readily suggests itself is that education makes people more likely to have physicalist intuitions. Perhaps, for example, it increases their confidence in the explanatory potential of the natural sciences. However, if understanding the "what it is like" locution as it was used in the survey required a level of linguistic competence, then it may be that it was better

understood by respondents with a higher level of education and that this accounts for the difference. Further research would need to decide between these hypotheses.

4.5. Limitations

The major limitations of the study were that the sampled population was the population of only one country (the United States) and that participating in the survey required English language skills. Further research on the prevalence of the Mary intuition – and whether it has population-level universality or something approaching this – would need to explore the rate that the intuition occurs (at least) in different countries and different linguistic communities.

5. Conclusion

The motivation for this study was to test the prevalence of the Mary intuition because of its relevance to the question of whether the intuitions at the heart of the mind-body problem are actually shared beyond the extremely small population of philosophers (or whether they might be largely idiosyncratic to that group and perhaps an artifact of their distinctive training) and because of its relevance to Chalmers' meta-problem of consciousness. In closing, it is also worthwhile to take a step outside this framework and simply to notice what the results tell us about the forcefulness of Jackson's thought experiment. That it elicits a particular intuition going to the heart of one of the great mysteries of philosophy and science – the mind-body problem itself – not only from most philosophers but also from such large majorities of the population across so many demographics is something quite remarkable.

Notes

1. This is more explicit in Jackson (1986).
2. We thank David Chalmers for emphasizing this to us.
3. And many sources since then; consult Nida-Rümelin and O'Conaill (2019) for a list.
4. There is a considerable literature on each of these responses. For discussion of the three responses and further references, see Nida-Rümelin and O'Conaill (2019).
5. Since then, a handful of studies have been carried out: Sytsma and Ozdemir (2019), Díaz (2021), and Fischer and Sytsma (2021).
6. Being UK-based, Prolific makes payments to participants in British currency, regardless of their location.
7. The exclusion criteria were decided upon after conducting the survey.
8. Examples are provided unedited.

9. On the face of it, anyone with the Mary intuition should also think that Mary would learn what it is like to see red when she wakes from the operation and sees the red tomato, because she will obviously know what it is like to see red after this. This might raise the question why we only expected a 'high correspondence' and not complete correspondence between the groups who had these intuitions. Primarily, we did not expect that all respondents would provide consistent responses, especially given that we were trying to source intuitions rather than considered positions, although we thought most would. It was also possible that some respondents would provide what appear to be inconsistent answers for sophisticated reasons.
10. We thank David Chalmers for this observation.
11. We thank an anonymous referee for this suggestion. The referee also suggests this elaboration. Respondents who reported the Mary intuition and who also thought that Mary had the ability to imagine red pre-operation, but who assumed that she did not exercise this ability, might be seen as (in effect) rejecting Nemirow's (1980) original version of the abilities response to the knowledge argument, or at least a weak version of it. Nemirow held that '[t]o understand an experience of seeing red . . . one must remember a similar experience, or (at least) be able to imagine one' (p. 475). Plausibly, the respondents with the combination of views just described were, in effect, denying that knowing what it is like to see red (or understanding such an experience) *only* requires the ability to imagine (because they thought that Mary did have this ability but did not know what it is like to see red). However, these respondents could not, without further data, be interpreted as (in effect) rejecting a stronger version of the abilities response on which knowing what it is like to see red requires not only the ability to imagine red but also, e.g., the ability to remember it or recognize it (see Lewis (1983) for the latter). It may be that they thought that Mary pre-operation 1) did not know what it is like to see red; 2) had the ability to imagine seeing red but did not exercise this ability; 3) did not have the additional abilities which are required to know what it is like to see red (though they might have gained them if she exercised her ability to imagine red).
12. We thank Krisztina Orbán for highlighting to us the importance of asking respondents not just if Mary would be able to imagine red pre-operation, but whether she would be able to do so *correctly*.
13. We thank an anonymous referee for drawing our attention to these issues. On the matter of whether laypeople have a concept of phenomenal consciousness, Chalmers (2018, pp. 13, Footnote 7) identifies the following sources as others among the relevant empirical work: Huebner (2010), Talbot (2012), and Peressini (2014).

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Data set

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